Assignment 4B – TalentWise Advisory

Section A		
#	**Questions**	**Answers**
1	Confusion matrix	True Negative: 4 False Positive: 10 False Negative: 5 True Positive: 53
2	Accuracy	79.16%
3	Precision	84.12%
4	Recall	91.37%
5	F1-score	87.60%
6	Training Accuracy	86.45%
7	Testing Accuracy	79.16%

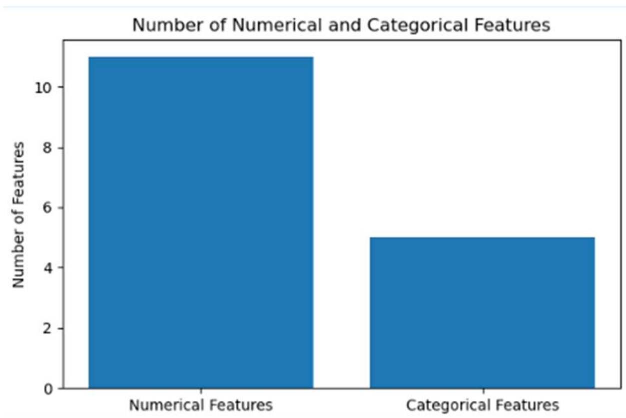
Section B		
#	**Questions**	**Answers**
1	What is the business about?	TalentWise Advisory is a mid-sized consulting and business services firm that provides project-based solutions to corporate clients. The company operates across multiple departments and relies on skilled employees, organizational knowledge, and strong team performance to deliver successful client outcomes.
2	What problem is the business trying to solve?	The business is trying to solve the problem of employee attrition (staff turnover) by identifying employees who are at risk of leaving before they resign.
3	What decision can machine learning help the business make?	Machine learning can help TalentWise identify employees who are likely to leave the company by analyzing key workforce factors such as engagement, workload, overtime, tenure, and job satisfaction. The model provides early warnings that enable HR and managers to take proactive retention actions, improve employee support, and reduce turnover, while leaving final decisions to human judgment.
4	What is the target variable in the dataset?	Left_Company
5	What are the input features?	Every column except Employee_ID and Left_Company
6	Why is this prediction useful for the business?	The prediction helps TalentWise proactively identify employees at risk of leaving, enabling timely retention efforts that reduce turnover costs, improve workforce stability, and maintain productivity and service quality.

Section C		
#	**Questions**	**Answers**
1	Then briefly explain whether the Decision Tree model may be overfitting	The Decision Tree model does not show significant signs of overfitting. The training accuracy is **86.46%**, while the testing accuracy is **79.17%**. The difference between the two scores is approximately **7.29 percentage points**, which is relatively small and indicates that the model generalizes reasonably well to unseen data.
2	How well did the Decision Tree model perform?	The Decision Tree model performed reasonably well in predicting employee attrition. Key evaluation results:

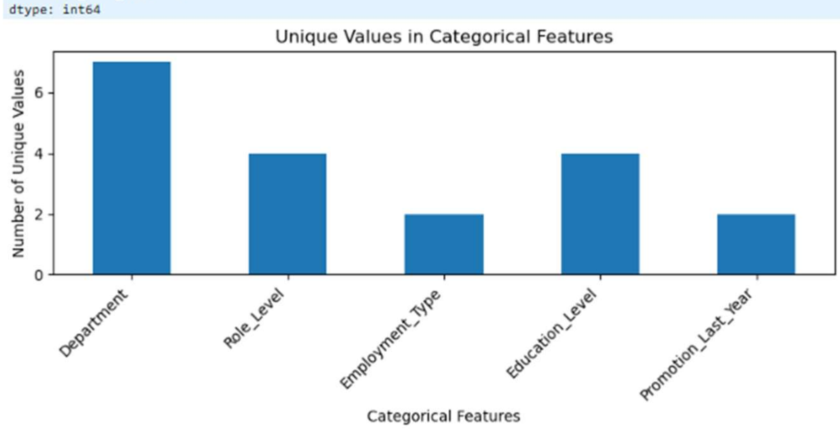
<table>
<tr><td></td><td></td><td><table><tr><td>Metric</td><td>Score</td></tr><tr><td>Accuracy</td><td>79.17%</td></tr><tr><td>Precision</td><td>84.13%</td></tr><tr><td>Recall</td><td>91.38%</td></tr><tr><td>F1-Score</td><td>87.60%</td></tr></table>The model correctly classified approximately 79% of employees in the testing dataset and achieved strong recall and F1-score values.</td></tr>
<tr><td>3</td><td>Is there a large difference between training accuracy and testing accuracy?</td><td>No, the difference is approximately 7.29%.
Training Accuracy = 86.46%
Testing Accuracy = 79.17%</td></tr>
<tr><td>4</td><td>Does the model show signs of overfitting? Why or why not?</td><td>The controlled Decision Tree model shows only minor signs of overfitting.
Evidence:
Training Accuracy = 86.46%
Testing Accuracy = 79.17%
The performance difference is not excessive.</td></tr>
<tr><td>5</td><td>Which evaluation metric is most important for this business problem?</td><td>Recall is the most important metric.
Recall measures how many employees who are actually likely to leave are correctly identified by the model.
The model achieved a Recall score of 91.38%, meaning it successfully identified most employees who were at risk of leaving.</td></tr>
<tr><td>6</td><td>What do false positives mean in this business context?</td><td>A False Positive occurs when:
The model predicts an employee will leave
The employee actually stays.
HR may spend time and resources on unnecessary retention efforts.
Such as, Retention bonuses, coaching, or engagement programs may be directed toward employees who were not planning to leave.
In this model, there were 10 False Positives.</td></tr>
<tr><td>7</td><td>What do false negatives mean in this business context?</td><td>A False Negative occurs when:
The model predicts an employee will stay.
The employee actually leaves.
Business impact:
In this context, the company loses employees unexpectedly and productivity may be disrupted. In this model, there were 5 False Negatives.
Since employee turnover is costly, False Negatives are often more serious than False Positives.</td></tr>
<tr><td>8</td><td>Which features were most important in the Decision Tree model?</td><td>Based on the business problem and the structure of the Decision Tree, the most influential features are likely to include:
Prior_Delays_Last_6M
Past_On_Time_Rate
Backorder_History
Inventory_Buffer_Days
Supplier_Rating
Promised_Lead_Time_Days
These features directly relate to supplier reliability, historical performance, and operational risk.</td></tr>
<tr><td>9</td><td>How can these important features help the business make better decisions?</td><td>By giving the Business better insight to their situation. This can help the Business / HR plan accordingly.</td></tr>
<tr><td>10</td><td>What is one possible limitation or bias in the model or dataset?</td><td>One limitation is the relatively small dataset size of 360 employee records.</td></tr>
</table>

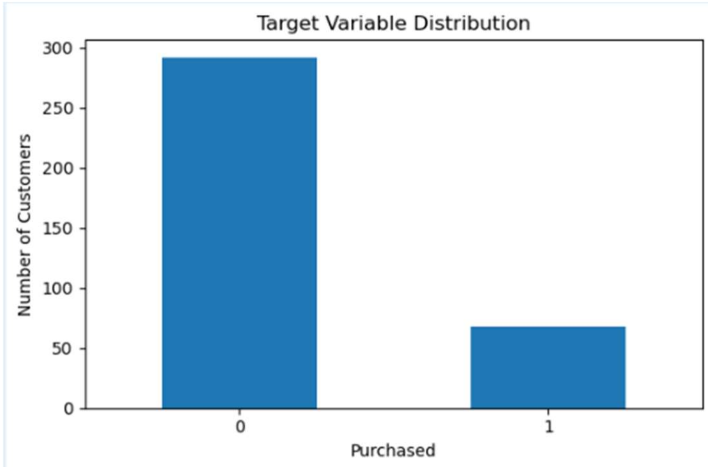
		A small dataset may not fully represent all employee behaviors, departments, or future workforce trends. There is also imbalance in the data set.
11	Why should human judgment still be used when making business decisions based on model results?	Machine learning models identify patterns from historical data but cannot understand every factor influencing employee behavior. Human judgment remains important because: • Employees may leave for personal reasons not captured in the dataset. • Organizational changes may affect future behavior differently. • Ethical and fairness considerations require human oversight. The model should therefore be used as a decision-support tool rather than a replacement for managerial judgment.

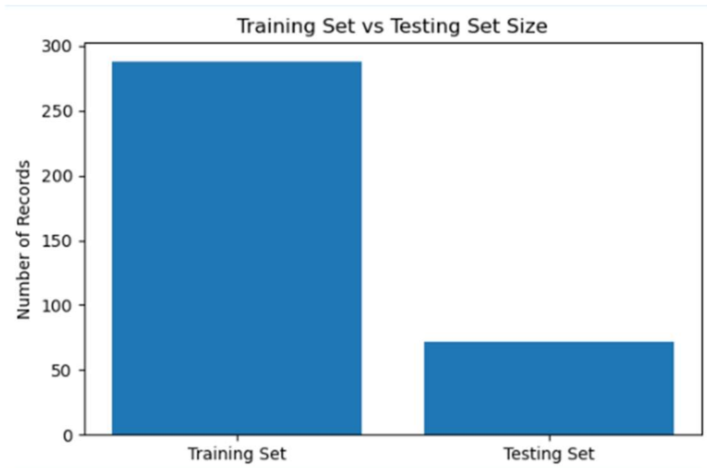
Graphical Charts / Representation

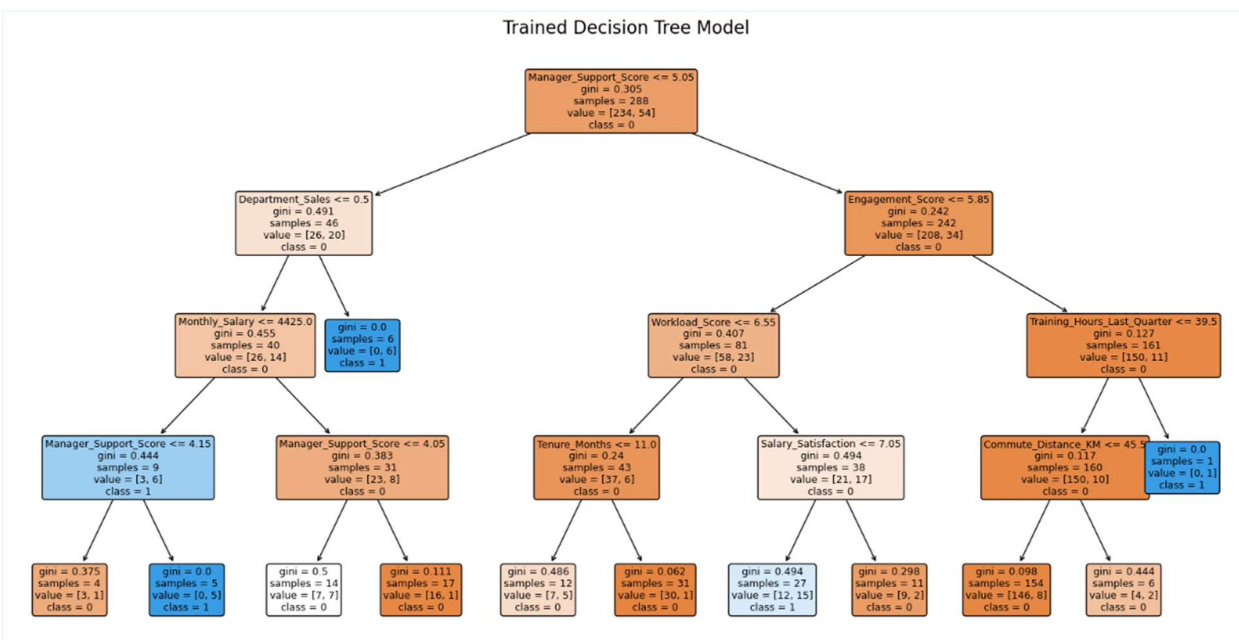



```
Number of numerical features: 11
Number of categorical features: 5
Number of unique values in each categorical feature:
---------------------------------------------------
Department           7
Role_Level           4
Employment_Type      2
Education_Level      4
Promotion_Last_Year  2
dtype: int64
```




Target Variable Distribution
Number of Customers
Purchased
0
1
0
50
100
150
200
250
300


Training Set vs Testing Set Size
Number of Records
Training Set
Testing Set
0
50
100
150
200
250
300


Trained Decision Tree Model
Manager_Support_Score <= 5.05
gini = 0.305
samples = 288
value = [234, 54]
class = 0
Department_Sales <= 0.5
gini = 0.491
samples = 46
value = [26, 20]
class = 0
Engagement_Score <= 5.85
gini = 0.242
samples = 242
value = [208, 34]
class = 0
Monthly_Salary <= 4425.0
gini = 0.455
samples = 40
value = [26, 14]
class = 0
gini = 0.0
samples = 6
value = [0, 6]
class = 1
Workload_Score <= 6.55
gini = 0.407
samples = 81
value = [58, 23]
class = 0
Training_Hours_Last_Quarter <= 39.5
gini = 0.127
samples = 161
value = [150, 11]
class = 0
Manager_Support_Score <= 4.15
gini = 0.444
samples = 9
value = [3, 6]
class = 1
Manager_Support_Score <= 4.05
gini = 0.383
samples = 31
value = [23, 8]
class = 0
Tenure_Months <= 11.0
gini = 0.24
samples = 43
value = [37, 6]
class = 0
Salary_Satisfaction <= 7.05
gini = 0.494
samples = 38
value = [21, 17]
class = 0
Commute_Distance_KM <= 45.5
gini = 0.117
samples = 160
value = [150, 10]
class = 0
gini = 0.0
samples = 1
value = [0, 1]
class = 1
gini = 0.375
samples = 4
value = [3, 1]
class = 0
gini = 0.0
samples = 5
value = [0, 5]
class = 1
gini = 0.5
samples = 14
value = [7, 7]
class = 0
gini = 0.111
samples = 17
value = [16, 1]
class = 0
gini = 0.486
samples = 12
value = [7, 5]
class = 0
gini = 0.062
samples = 31
value = [30, 1]
class = 0
gini = 0.494
samples = 27
value = [12, 15]
class = 1
gini = 0.298
samples = 11
value = [9, 2]
class = 0
gini = 0.098
samples = 154
value = [146, 8]
class = 0
gini = 0.444
samples = 6
value = [4, 2]
class = 0

	Predicted No	Predicted Yes
Actual No	4	10
Actual Yes	5	53